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Investor attention and global market returns during the COVID-19 crisis

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A B S T R A C T

The financial market response to the COVID-19 pandemic provides the first example of a market crash instigated by a health crisis. As such, the crisis provides a unique setting in which to examine the market response to changes in investor attention. We utilise Google search volume (*GSV*) as a proxy for investor attention. *GSV* for the "coronavirus" keyword increases markedly from late-February and peaks in mid-March before declining substantially. Our results are broadly consistent with Da, Engelberg, and Gao (2015), indicating that *GSV* is primarily a proxy for the attention of retail investors and confirming that investor attention negatively influences global stock returns during this crisis period. A rise in the number of internet searches during the COVID-19 crisis induces a faster rate of information flow into financial markets and so is also associated with higher volatility. The identified relationships are economically and statistically significant even after controlling for the number of COVID-19 cases and macroeconomic effects. Increases in *GSV* have less impact on government bond yields where the limited role of *GSV* is likely due to lower participation of retail investors. The results suggest that, rather than searching for information on potential stocks to buy (Barber & Odean, 2008), retail investors are searching for information to resolve uncertainty about household FEARS (Da et al., 2015) during the COVID-19 crisis.

result

1. Introduction

Throughout history, investors have faced the prospect of financial market crisis. Typically, financial crises have been precipitated by economic crises that result from pro-cyclical changes in the supply of credit (Kindleberger, 1978). The COVID-19 pandemic provides the first example of a market crash instigated by a health crisis. Starting with the Chinese decision to isolate the city of Wuhan on 23 January, through the late February surge of cases in Iran and Italy, to the mid-March US declaration of a national emergency,[1] global stock markets responded negatively to unfolding events (Fig. 1). From peak-to-trough, the average G7 stock market fell 36.3% (similar to the 35.4% fall across the G20).

The strong negative shock to market returns (Zhang, Hu, & Ji, 2020) and liquidity (Baig, Butt, Haroon, & Rizvi, 2020) is linked to the growth in confirmed cases of COVID-19 (Al-Awadhi, Alsaifi, Al-Awadhi, & Alhammadi, 2020), and had a greater impact on firms with exposure to China (Ramelli & Wagner, 2020). Possibly due to the increasingly rapid diffusion of news, no prior episode of viral outbreak has led to a market reaction that remotely resembles the response to COVID-19, and the frequency of stock market jumps is greater than at any other period in history (Baker et al., 2020). Under such extreme conditions, it is evident that some financial instruments act as amplifiers, rather than hedges, of risk (Corbet, Larkin, & Lucey, 2020).

We attempt to explain the direction and magnitude of the market response through the lens of *investor attention*. When investors are more attentive to news events, information is more quickly incorporated into prices, inducing a price response and higher return volatility. Indeed, prices only respond to new information if investors pay attention to it (Huberman & Regev, 2001). A larger risk premium is then required by investors to account for the additional attention-induced risk (Andrei & Hasler, 2014). The issue of cognitive constraint (Kahneman, 1973) on investor attention is increasingly important in the era of social media and the vast amount of information related to the COVID-19 pandemic. Essentially, one could think of "investor attention" (our focus) as awareness as to whether a piece of information exists, whereas "investor sentiment" is the mood-biased interpretation of the information as being favourable or not for asset prices. The terms are related to the extent suggested by Lim and Teoh (2010), whereby greater investor attention can exacerbate the effect of investor behavioural biases (such as sentiment) on asset prices.

The literature provides two competing theories to explain the market

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[1] For a timeline of the COVID-19 crisis see https://www.nytimes.com/article/coronavirus-timeline.html

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1057-5219/

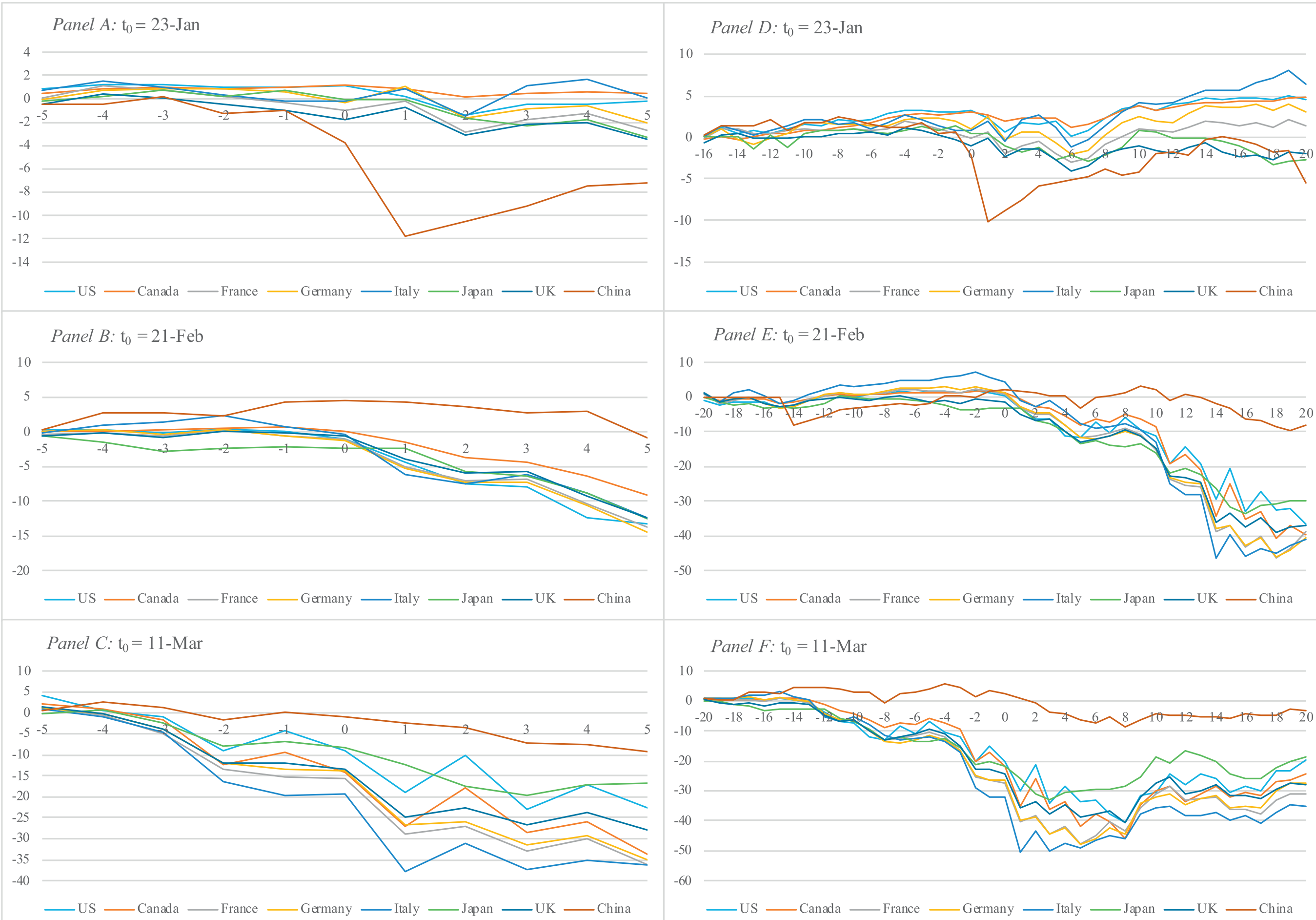

Fig. 1. Cumulative G7 stock returns across various event windows.

response to investor attention, both incorporate the idea that attention is a scarce resource (Kahneman, 1973). Odean (1999) and Barber and Odean (2008) present a theory based on *choice asymmetry*. This postulates that the possible investment universe is extremely large when investors are buying stocks and so only those stocks that grab attention are purchased. Conversely, only stocks already held are typically sold and so the same search problem does not exist. Thus, increased attention leads to increased buying and temporarily higher prices that are subsequently reversed but has no impact on selling frequency.

The alternate theory argues that increased investor attention leads to greater *information discovery* (Tantaopas, Padungsaksawasdi, & Treepongkaruna, 2016; Vlastakis & Markellos, 2012). This generates temporary price pressure but also improves market efficiency by reducing return predictability (Vozlyublennaia, 2014). The notion that increased levels of attention may increase the amount of information that flows into the market, particularly during the COVID-19 crisis, relates to the mixture of distributions hypothesis posited by Smith (2012) as an explanation for the volatility response. Our empirical results are supportive of the information discovery explanation.

We use Google search volume (*GSV*) as our proxy for investor attention. This measure appeals for several reasons. The internet provides a vast trove of freely available information, it is easily accessible around the globe, and usage has grown exponentially since the mid-1990s. Over 80% of the population in developed countries, and 50% globally, use the internet.[2] Google dominates the search engine market with approximately 90% of global search query volume[3] and so Google search frequency is likely to be representative of the search behaviour of the general population. *GSV* for the "coronavirus" keyword increases substantially from late-February and peaks in mid-March (as stock markets bottom and the number of COVID cases initially peaks) before declining substantially (as stock markets recover).

Performing an internet search query is a *direct* indication of attention. Searching for information about a particular subject clearly shows that one is paying attention to that subject. Thus, *GSV* provides a direct and timely measure of the acquisition of available information. Choi and Varian (2012) suggest that *GSV* the potential to "nowcast" a variety of economic activities and, somewhat appropriately given the current study, it is even used to predict influenza outbreaks (Ginsberg et al., 2009).

Since institutional investors have access to a variety of professional news sources, such as Bloomberg and Reuters, and do not have to rely on Google searches, our measure of investor attention primarily relates to *retail* investors. This is consistent with prior studies that have utilised *GSV* to proxy attention of households (Da et al., 2015; Kostopoulos, Meyer, & Uhr, 2020), retail investors (Ding & Hou, 2015; Yung & Nafar, 2017), and uninformed traders (Bank et al., 2011). Ben-Rephael, Da, and Israelsen (2017) provide evidence that readership of Bloomberg articles is a good proxy for institutional but not retail investors. This would also be consistent with our empirical results indicating *GSV* has a limited influence on markets with lower retail participation (Japanese stocks

[2] https://ourworldindata.org/internet

[3] https://au.oberlo.com/statistics/search-engine-market-share

and G7 government bond yields).

Several other studies have utilised *GSV* as a proxy of investor attention. The evidence suggests that investor attention varies over time and impacts asset prices (Da, Engelberg, & Gao, 2011). Our paper has similarities to that of Da et al. (2015), a although they construct a Financial and Economic Attitudes Revealed by Search (*FEARS*) index, using *GSV* for a number of economic related search terms, while we focus on a single search term. Consistent with the over-reaction associated with sentiment-induced mispricing, they find that an increase in FEARS is related to low returns today followed by a reversal in the following days. The reversal pattern is also noted by Tang and Zhu (2017) and Cheng, Huang, and Hu (2019) who show that reversals occur when price shocks are not accompanied by new information and are stronger for stocks that receive less attention. Heyman, Lescrauwaet, and Stieperaere (2019) are able to profit from this pattern by generating a profitable trading strategy by selling winner stocks that experience an increase in *GSV*.

An increase in Google searches is also related to higher levels of stock market liquidity (Bank et al., 2011; Ding & Hou, 2015), trading volume, and volatility (Vlastakis & Markellos, 2012; Aouadi et al., 2013; Dimpfl & Jank, 2015), and explains the speed with which stock prices respond to analyst recommendations (Welagedara, Deb, & Singh, 2017). Investor attention, measured by *GSV*, also explains high initial returns to IPOs (Vakrman & Kristoufek, 2015), and market responses in commodity markets (Han, Li, & Yin, 2017a; Han, Lv, & Yin, 2017b), cryptocurrency markets (Urquhart, 2018), FX markets (Goddard, Kita, & Wang, 2015; Han et al., 2018; Smith, 2012), and REITs (Yung & Nafar, 2017).

The literature has not fully resolved the direction of the relationship between *GSV* and stock returns. For instance, Bijl, Kringhaug, Molnar, and Sandvik (2016) and Chen (2017) identify a negative relationship whereby returns decline following an increase in investor attention, contrarily Da et al. (2011) and Tang and Zhu (2017) find a positive relationship, while Kim, Lucivjanska, Molnar, and Villa (2019) find no significant relationship at all. We add to the literature by attempting to empirically resolve this tension and our results provide support for the negative association. Understanding these dynamics is particularly important in light of warnings by regulators as to the additional risks facing retail investors[4] when trading during periods of extreme volatility.

Our initial set of empirical tests adopt a methodology similar to Da et al. (2015). We demonstrate that, except for Japan,[5] an increase in investor attention towards the "coronavirus" search term results in lower stock returns across the major G7 stock indexes. The identified effect is economically and statistically significant and remains even after controlling for the number of COVID-19 cases and various macroeconomic effects. There is a statistically insignificant reversal on the day following an increase in investor attention, but the initial shock lasts for up to 4 trading days. We also find a negative relationship in G20 stock returns, but aside from Australia, it is only significant when using global *GSV*. This may be due to a low proportion of market participation from domestic retail investors in emerging markets.

Our second set of tests applies an EGARCH framework to examine the effect of *GSV* on the conditional volatility of stock returns and changes in government bond yields. Consistent with Vlastakis and Markellos (2012), Aouadi et al. (2013), and Dimpfl and Jank (2015), we find that investor attention is positively related to conditional volatility. This suggests that a rise in the number of internet searches during the COVID-19 crisis induces a faster rate of information flow into financial markets. By focusing on "pandemic" search interest, we find some evidence that our results also hold over the longer term. The last section of empirical testing focuses on changes in government bond yields. While investor attention positively impacts conditional volatility of bond yields across the G7, the mean response is only well defined for Italian government bonds, where it is positive. This would seem to be consistent with the particularly severe additional strain that the COVID-19 outbreak has placed on Italian government finances.[6]

Broadly speaking, our results are consistent with Da et al. (2015), indicating that *GSV* is primarily a proxy for the attention of retail investors, confirming that investor attention negatively influences stock returns, and is associated with higher volatility. The results suggest that, rather than searching for information on potential stocks to buy (Barber & Odean, 2008), retail investors are searching for information to resolve uncertainty about household FEARS (Da et al., 2015). They are also consistent with greater investor attention exacerbating the effect of investor behaviour (Lim & Teoh, 2010).

We proceed as follows. Section 2 provides an overview of the data used in the study with a focus on market returns and our proxy for investor attention. Section 3 provides empirical analysis together with discussion, and section 4 concludes.

2. Data

2.1. Google search volume

The Google search volume (*GSV*) index, obtained via Google Trends,[7] provides a time series of the volume of search queries and is utilised as our measure of investor attention. *GSV* normalizes data on search queries so that numbers are scaled between zero and 100 based on a topic's proportion to all searches on all topics within a particular geography. A value of 100 would then indicate the particular search term is extremely active for the selected time frame and location. Google Trends applies filters to remove duplicate searches, searches with special characters, and searches made by very few people. The index is particular to a specific sample period. As the length of the sample period increases, the frequency of data provided by Google Trends decreases. For example, daily data is available for periods up to 3-months, weekly data for periods up to 5-years, and monthly data for longer intervals. Prior work which has sought to employ daily data over long sample periods (e.g. Bijl et al., 2016; Da et al., 2011, 2015) have had to normalise the raw data in some way. The relatively short sample period of our study avoids this issue and so no scaling is required.

Our study primarily focuses on searches for the "coronavirus" keyword.[8] This keyword is chosen as it is specific to our sample period (unlike "epidemic" or "pandemic") and circumstance (unlike "virus" which may also relate to computers). In addition, there is also evidence that this term received large amounts of attention outside of Google.[9] Fig. 2 depicts the trend for "coronavirus" search term during our sample period across different geographies. Coinciding with the lockdown of Hubei province, the term received increased attention from 23-January onwards (particularly in China). Attention then declined before rising again at the end of February as a significant rise in COVID-19 cases were reported in Iran and Italy. The index peaks in mid-March as it became evident that the virus had impacted large swathes of Europe and the US, and the US declared a national emergency. The global index then falls towards the end of the sample period, with several spikes related to virus outbreaks in specific countries (e.g. Japan in early April and mid-June).

[4] https://asic.gov.au/about-asic/news-centre/find-a-media-release/2020-releases/20-102mr-retail-investors-at-risk-in-volatile-markets/

[5] The estimated results for Japan are in the same direction as the other countries but statistically insignificant.

[6] https://www.washingtonpost.com/business/2020/03/11/financial-crisis-coronavirus-italy/

[7] https://trends.google.com/

[8] We repeat our analysis using alternate search terms – such as "COVID" and "COVID-19" – which are less commonly searched than "Coronavirus" and report results as part of our robustness testing (see Table 9).

[9] https://www.economist.com/graphic-detail/2020/03/25/the-language-of-covid-19-has-people-turning-to-the-dictionary

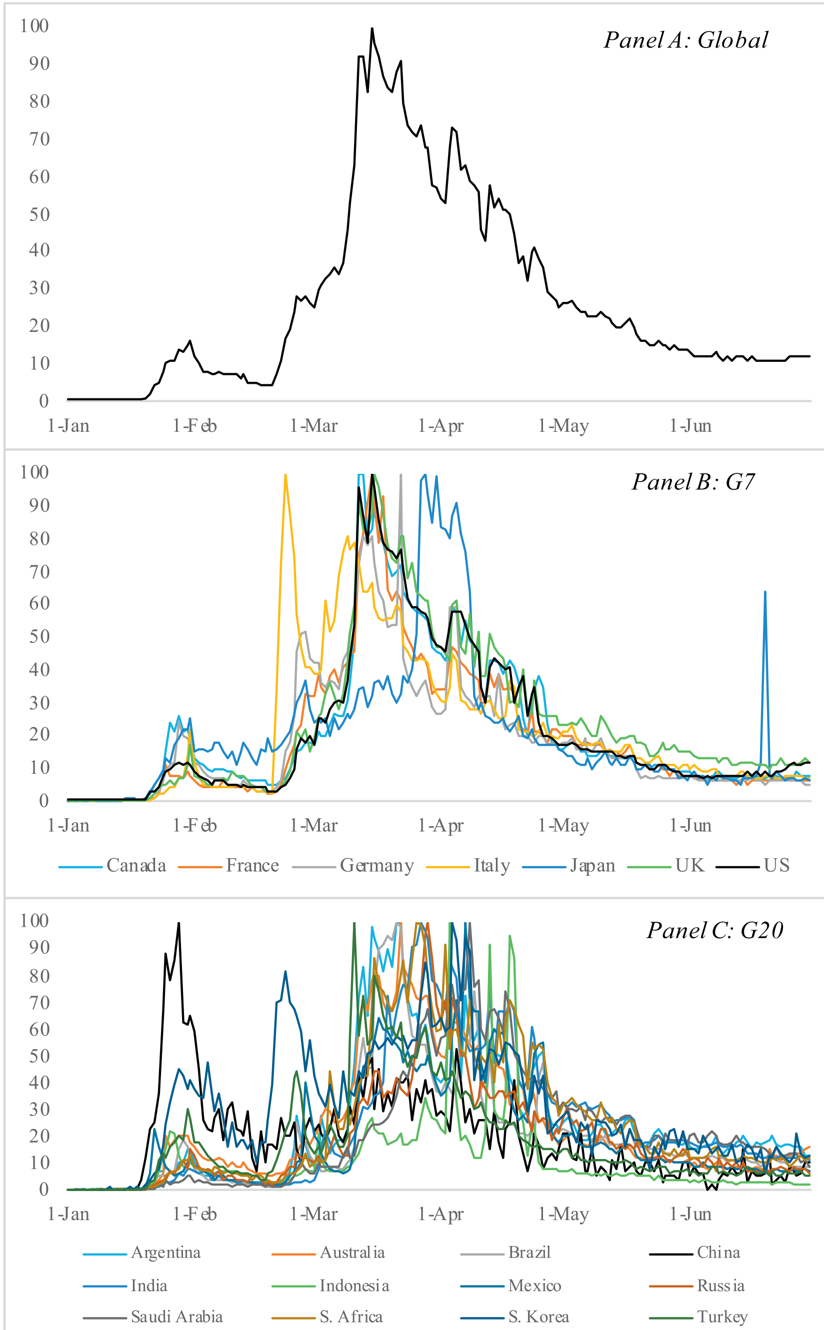

Fig. 2. Google search volume for "Coronavirus" search term (by country).

Descriptive statistics for "coronavirus" *GSV* index are shown in Table 1. Except for Indonesia (with a lower mean *GSV* of 10.33), the attention given to "coronavirus" is similar across the G20 (averaging 21.8), and the *GSV* of individual countries is highly correlated with global *GSV*. The explanatory variable used in our study is the daily change (first-difference) in *GSV* and there is no evidence of a unit root in this time series.

It is possible that in addition to *GSV*, investors are also influenced by news directly related to the spread of the coronavirus. We therefore control for the number of newly reported COVID-19 cases (the number of *new* cases may also be interpreted as the daily *change in total* cases reported) using information provided by the European Centre for Disease Prevention and Control (ECDC).[10] The ECDC collects and harmonizes data related to the pandemic from around the world, providing a global perspective.[11] During our sample period the global daily average of new COVID-19 cases was over 55,000. Fig. 3 shows that the number of cases initially peaked in April 2020, started to decline, and then

[10] https://www.ecdc.europa.eu/en/publications-data/download-todays-data-geographic-distribution-covid-19-cases-worldwide

[11] The data is used by respected resources such as the "Our World in Data" website supported by the University of Oxford: https://ourworldindata.org/coronavirus-source-data

Table 1
Descriptive Statistics: Google Search Volume (GSV) for "Coronavirus".

	Mean	Std. Dev.	Corr_w_Global
Global	26.81	25.20	1.00
G7			
Canada	21.60	22.19	0.97
France	19.38	20.43	0.96
Germany	18.60	18.27	0.89
Italy	21.03	21.10	0.77
Japan	20.91	20.96	0.69
UK	24.34	23.47	0.98
US	21.89	23.05	0.99
G20			
Argentina	24.50	23.28	0.69
Australia	23.30	22.31	0.97
Brazil	19.85	21.97	0.90
China	18.76	17.27	0.39
India	24.39	24.75	0.80
Indonesia	10.33	13.63	0.60
Mexico	19.20	20.10	0.85
Russia	18.12	17.44	0.83
Saudi Arabia	21.73	21.61	0.67
South Africa	25.31	25.38	0.89
South Korea	29.88	21.29	0.73
Turkey	18.88	19.66	0.90

increased again from the end of May primarily due to further outbreaks in the US, Brazil and India.

2.2. Financial asset returns

We follow Tantaopas et al. (2016) in focusing on the effect of investor attention at the index level. In the first instance, we consider the leading stock indexes across G7 nations and then expand the sample to include the stock indexes of the remaining G20 nations.[12] As a supplementary test we then examine whether investor attention impacts G7 government bond yields. Together, the G20 economies constitute approximately 90% of global GDP and their stock exchanges represent over \$58 trillion of market capitalization (with the G7 accounting for around 40% and \$46 trillion respectively). We obtain data from DataStream at a daily frequency and use this to compute a time series of daily stock returns (where $r_t = \ln(index_t/\ index_{t-1})$). We subtract the risk-free rate to obtain the excess of log returns. For bond yields, the presence of a mixture of negative and positive interest rates means that we are restricted to using first differences.

Fig. 4 shows that there was a sharp decline in G20 stock markets from late February to mid-March 2020 and that a portion of this was subsequently reversed prior to the end of the sample period in June 2020. For the US, Canada, Germany, and Japan the reversal brought the index almost back to pre-pandemic levels. The story for 10-year bond yields (Panel B) is more nuanced with yields initially falling, then rising from mid-March as central banks (particularly the US Federal Reserve) announced market interventions.[13] With the exception of Italy, rates then resumed a downward trend until the end of the sample period. An initial indication of the effect on market volatility is found by looking at squared daily returns (not shown) where there is a sharp increase that is concentrated in the period 09-March to 26-March.

Table 2 provides descriptive statistics for the stock markets and government bond yields of interest. Unsurprisingly, the mean stock return and bond yield change is negative during the sample period (with the exception of Japanese yields). The UK stock market index (FTSE) suffers the sharpest mean daily fall (−0.166%) in the G7 and the Indonesian market (IDX) has the lowest average fall in the G20 (−0.200%). The Italian (MIB) and Brazilian (Bovespa) indexes are the most volatile in the G7 and G20 with standard deviations of 2.94% and 3.98% respectively. The greatest daily change in bond yields arises in the US while Italian bonds have the highest standard deviation (around 50% greater than the next most). All returns show evidence of a high degree of kurtosis (or "fat-tails") that is commonly found in financial assets, and there is no evidence of unit roots when testing using Augmented Dickey Fuller with intercept and trend.

3. Empirical analysis

3.1. Methodology

We begin our empirical analysis by examining the relationship between "coronavirus" *GSV* and stock returns. We adopt a methodology similar to that of Da et al. (2011, 2015) and Han et al. (2017a).

$$r_{t+k} = \beta_0 + \beta_1 \Delta GSV_t + \beta_2 log(Cases_t) + \beta_3 r_{t-1} + \sum_m \gamma_m Control_{m,t} + \varepsilon_{t+k} \ (1A)$$

$$r_{t+k} = \beta_0 + \beta_1 \Delta GSV_t + \beta_2 log(Cases_t) + \beta_3 r_{t-1} + \varepsilon_{t+k} \ (1B)$$

Where r_t denote the stock market return on day t, and r_{t-1} is the lagged return. ΔGSV_t is the daily change in the Google search volume of the "coronavirus" key word, for a specific country or at the global level. Log ($Cases_t$) is the natural log of the number of new COVID-19 cases[14] reported for day t (we add one to prevent an error in computing the log of 0 on days when no new cases are reported). When using the country-specific *GSV* we also use the country-specific number of *cases*, and when using the global *GSV* we use the global number of *cases*. For the US, we are able to incorporate a set of control variables (Eq. 1A) to control for macroeconomic conditions and ascertain the incremental impact of investor attention. This set of variables includes daily changes in a news-based measure of economic policy uncertainty (*ΔEPU*), changes in the Aruoba-Diebold-Scotti *(ΔADS)* business conditions index, and changes in the Treasury bill – Eurodollar futures (*TED*) spread. Unfortunately, similar measures are not available at a daily frequency for the other countries in the sample and so in the majority of cases we apply Eq. (1B). ε_t are Newey-West standard errors.

3.2. GSV and G7 stock returns

Before addressing the effect of investor attention in the broader G7 or G20 markets, we assess the impact on the largest stock market, the US. Table 3 reports the estimated coefficients from eq. (1A) where the dependent variable is the return on the S&P500. The first column is our baseline regression. A highly significant and negative contemporaneous relationship is identified between S&P500 returns and *GSV* changes. In other words, returns are negative when investors place more attention on searches of the "coronavirus" key word. A 1 standard deviation increase in *GSV* is associated with a 5.49% decline in the S&P500 index (−0.251 × 21.89). While this sign is in the opposite direction to that found elsewhere in the literature (e.g. Tang & Zhu, 2017) it is consistent with that of Da et al. (2015). This reflects the fact that the search term that investors are placing additional attention towards has negative implications for asset prices (as is the case with the FEARS of Da et al. (2015)).

The estimated coefficients for the number of COVID-19 cases is statistically insignificant (this is true for each country throughout our

[12] G7 includes Canada, France, Germany, Italy, Japan, UK, and US. G20 includes G7 plus Argentina, Australia, Brazil, China, India, Indonesia, Mexico, Russia, Saudi Arabia, South Africa, South Korea, and Turkey.

[13] See for example: https://www.federalreserve.gov/newsevents/pressreleases/monetary20200315a.htm

[14] We do not also include the number of COVID-19 related deaths because they are highly correlated (~0.9) with the number of cases and so would introduce an element of multi-collinearity.

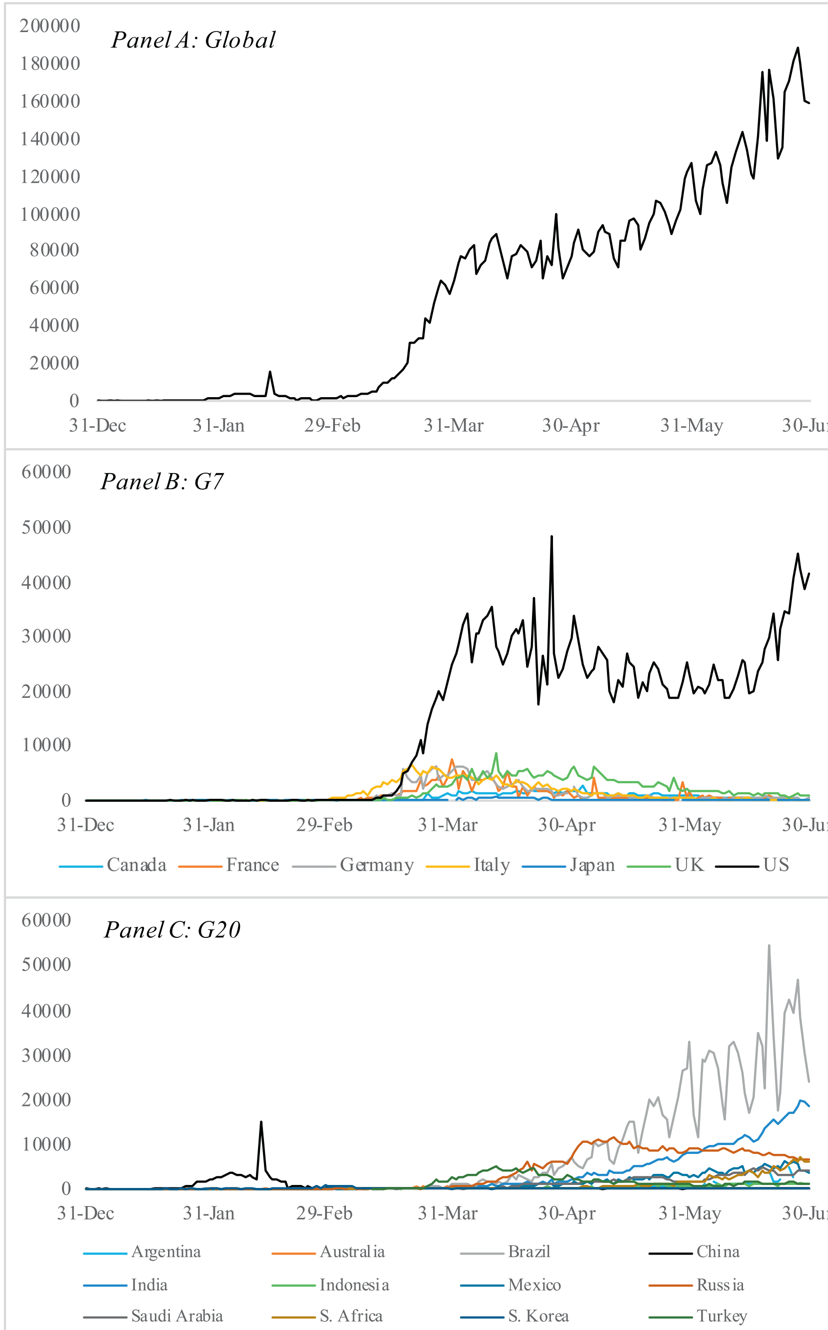

Fig. 3. Daily reported number of new cases of "Cornavirus" (COVID-19).

analysis). The signs of the coefficients for the control variables are consistent with ex-ante expectations, such that lower stock returns are associated with higher levels of economic uncertainty (*EPU*), worsening business conditions (*ADS*), and worsening credit conditions (*TED*). While the signs are in the expected direction, only the *EPU* coefficient is statistically significant. There is also evidence of negative autocorrelation of returns.

The 2nd column introduces lags of *GSV*, the investor attention variable. The contemporaneous *GSV* coefficient remains well-defined but the lags are not statistically significant. Column 3 demonstrate the limited explanatory power of our model without *ΔGSV*. In columns 4–7 we follow the methodology of Da et al. (2015) in assessing the longer-term impact of investor attention on stock returns. As with Da et al., we find that the returns are partially reversed at $t+1$ ($k=1$). Unlike Da et al., this reversal is not statistically significant, and there then appears to be a continuation of the initial response for $k=2$. The effect of *GSV* seems to be concentrated in the first three days. We find that this pattern, whereby a strong initial response to investor attention is followed by a partial reversal and is concentrated in the first 3-days or less, is common for the other markets and so we focus only on the contemporaneous response in other tables.

We next expand our consideration set to include the other G7 stock markets. As the set of control variable is not available for these markets we rely on Eq. (1B) for our analysis and report the results in Table 4. To be consistent across markets, we also rerun the regression for S&P500 using this regression model. For the sake of brevity, we report only the estimated *GSV* coefficient for each market. In panel A, the *GSV* time series used is for the "coronavirus" search term within the domestic

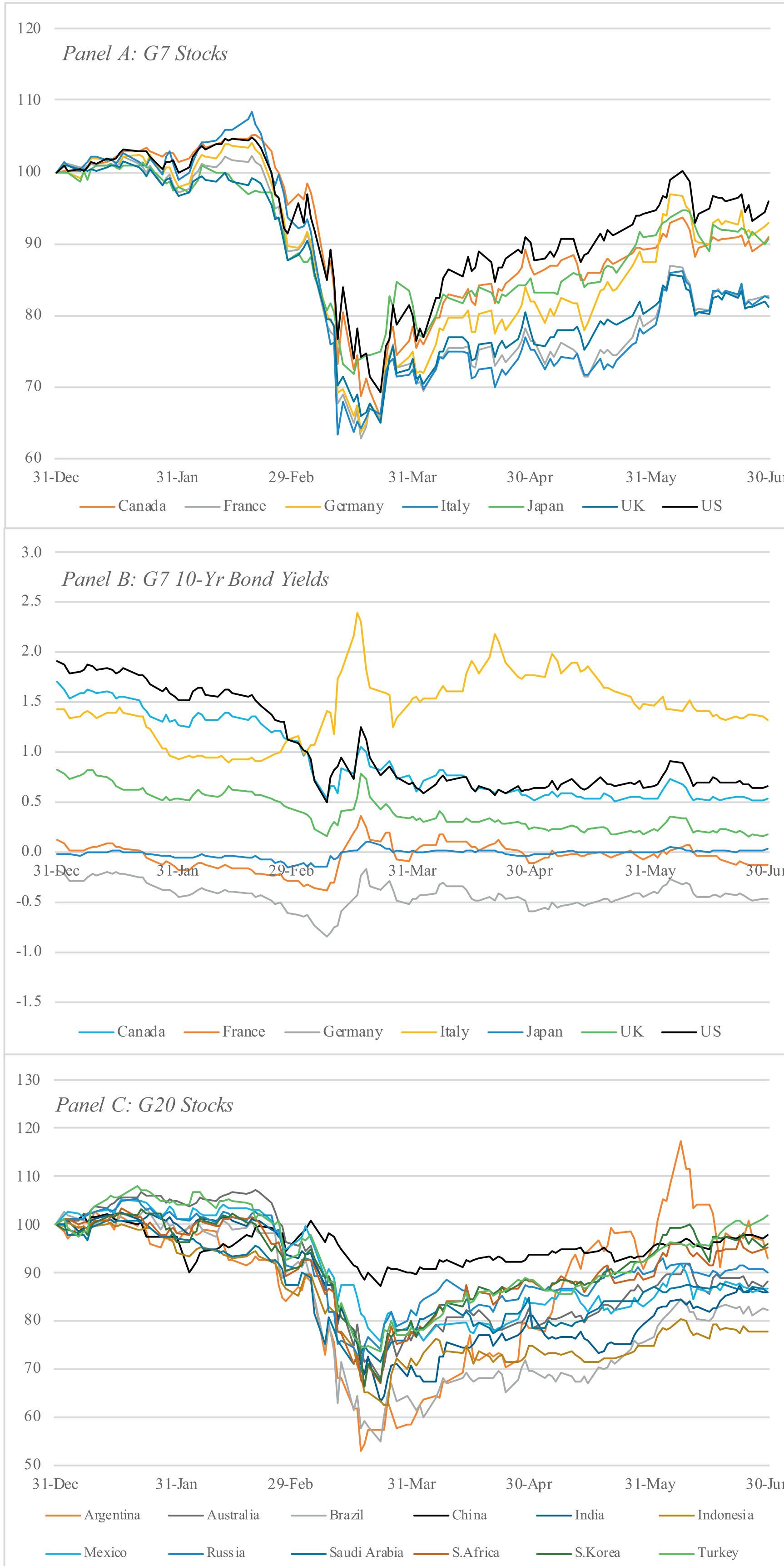

Fig. 4. Stock market performance and government bond yields (H12020).

Table 2
Descriptive Statistics: Stock Markets and 10-Year Government Bonds.

		Level	Returns			
Panel A: G7 Stock Markets		Mean	Mean	Std. Dev.	Skew	Kurtosis
Canada	*TSX*	15,488.5	−0.076	2.89	−0.99	10.16
France	*CAC*	5038.7	−0.153	2.63	−1.15	7.93
Germany	*DAX*	11,706.4	−0.059	2.66	−0.80	8.60
Italy	*MIB*	19,890.9	−0.155	2.94	−2.27	16.19
Japan	*TOPIX*	1540.5	−0.079	1.75	0.15	5.34
UK	*FTSE*	3550.8	−0.166	2.32	−0.88	7.76
US	*SP500*	2995.2	−0.033	2.93	−0.38	7.14
Panel B: G20 Stock Markets						
Argentina	*Merval*	36,383.4	−0.059	3.87	−0.75	5.76
Australia	*ASX*	6006.3	−0.100	2.50	−0.80	5.97
Brazil	*Bovespa*	94,005.1	−0.157	3.98	−1.09	8.34
China	*SSE*	3049.1	−0.017	1.34	−1.96	12.82
India	*NIFTY*	8456.6	−0.122	2.49	−1.59	10.74
Indonesia	*IDX*	5153.8	−0.200	2.01	0.53	7.95
Mexico	*Bolsa*	39,211.4	−0.114	1.92	−0.42	4.00
Russia	*MOEX*	2777.2	−0.083	2.10	−0.69	7.99
Saudi Arabia	*Tadawul*	7271.5	−0.120	1.97	−1.41	9.33
South Africa	*JSE*	51,818.4	−0.039	2.47	−0.92	6.50
South Korea	*KOSPI*	2015.8	−0.033	2.26	−0.03	6.16
Turkey	*BIST*	106,715.4	−0.015	1.88	−1.28	8.22
Canada		0.900	−0.942	6.03	0.42	6.14
France		−0.041	−0.193	4.93	0.65	6.95
Germany		−0.423	−0.218	4.85	0.48	6.50
Italy		1.448	−0.080	11.07	0.62	11.00
Japan		−0.014	0.040	1.83	1.66	12.61
UK		0.401	−0.522	4.76	0.95	8.79
US		1.026	−1.006	7.50	0.42	6.66

geographical region related to the specific stock market. For instance, the volume of Google searches for “coronavirus” in Canada is the dependent variable for the TSX returns. In Panel B, we use the global *GSV* time series, reflecting the worldwide volume of searches, and the global number of new COVID-19 cases.

The estimated coefficients show that stock returns are negatively related to investor attention on “coronavirus” across the G7. This relationship is statistically significant for both domestic and global *GSV* (Japan domestic *GSV* is the exception). The effect of a 1 standard deviation increase in domestic *GSV* ranges from −0.33% (Japan) to −5.58% (US) and for global *GSV* ranges from −2.14% (Japan) to −10.03% (Italy). The explanatory power also appears to be relatively high with up to 34% of the proportion of variance explained by modelling both domestic and global *GSV*. The number of COVID-19 cases (not reported) does not have a significant impact on stock returns in *any* G7 country.

The small effect of domestic *GSV* on Japanese stock returns may be related to the small proportion of trading volume conducted by retail investors on the Tokyo stock exchange.[15] For instance, in April 2020 just 0.165% of sales and 0.175% of purchases were on behalf of retail investors. Since institutional investors have access to a wider range of news sources it is likely they will rely on Google searches for information. Similarly, the small but relatively larger effect of global *GSV* for Japan may be related to the large proportion of foreign investors (73%).

3.3. GSV and G20 stock returns

In Table 5, we examine whether the stock markets within the rest of the G20 are similarly influenced by investor attention. We find that, except for Australia, the influence of domestic *GSV* is much smaller than for G7 nations and not statistically significant (Panel A). Australia may be an exception owing to the relatively high proportion of retail investors, and the sharp increase in their trading activities during the crisis period. An increase so substantial that it prompted a warning from the market regulator.[16]

In Panel B, we find that the impact of global *GSV* is much larger. The identified relationship is again negative, and the relationship is statistically significant for 9 of the 12 countries. The effect of a 1 standard deviation increase in global *GSV* ranges from −0.46% in China (where the COVID-19 outbreak occurred early and seemed to be well controlled by the end of our sample period) to −9.9% in Brazil (where the number of COVID-19 cases continued to increase through the end of our sample period). Brazil is also the only country in which the number of COVID-19 cases (not reported) has a statistically significant impact on market returns.

There are several possible explanations for the lighter influence of Google searches on some of the G20 stock markets. First, if *GSV* is a reflection of retail investor attention, the results may reflect a lower proportion of retail investors in those countries. Second, the search term used in our study may not be representative of the internet searches in some countries owing to language differences. For instance, “coronavirus” is also “coronavirus” in Portuguese (Brazil) and Spanish (Mexico) but translates to “koronaavairas” in Hindi (India). Third, *domestic GSV* in each country is more closely linked with *local* cases and lockdowns as opposed to the *global* outbreak. If stock markets, particularly those in emerging markets, are more focused on global issues then *global GSV* is likely to be more important. Finally, the proportion of internet searches conducted using Google differs across countries. In many countries Google handles over 95% of search volume[17] with Turkey (85%), Russia (50%), and China (2%[18]) the only exceptions in our sample. Indeed, the very low usage of Google in China would explain why the estimated *global* ΔGSV coefficient is the smallest by a large magnitude.

3.4. Robustness tests

Before moving on to examine the relationship between GSV and conditional volatility, we perform several robustness tests on our initial analysis. First, Da et al. (2015) form their FEARS index after winsorizing, deseasonalizing, and standardizing the 118 search terms that constitute the index. This allows for comparison across the 118 different search terms. Since we focus on a specific search term that does not appear to demonstrate seasonality we did not follow this process. However, it is still possible that extreme outliers could influence our reported results and that some element of seasonality may exist owing to non-observable patterns in *GSV*. We therefore repeat our earlier analysis once the ΔGSV time-series is adjusted following Da et al. (2015). First, we winsorize at the 5% level (2.5% in each tail). To address seasonality, we regress ΔGSV on weekday and month dummies and keep the residual. Finally, we standardize by scaling using the standard-deviation of ΔGSV.

Panel A of Table 6 reports the estimated coefficients for G7 countries using winsorized, deseasonalized, and standardized global ΔGSV. For all countries, the sign and statistical significance of the estimated coefficients is the same as previously reported (Table 4). However, the magnitude of the association is smaller than earlier indicated. For instance, when interpreting Table 4, we suggested that the US stock

15 https://www.jpx.co.jp/english/markets/statistics-equities/investor-type/00-01.html
16 https://asic.gov.au/about-asic/news-centre/find-a-media-release/2020-releases/20-102mr-retail-investors-at-risk-in-volatile-markets/
17 Source: www.gs.statcounter.com
18 Google is only available via virtual private network (VPN) in mainland China.

Table 3
Regression: relationship between "Coronavirus" search volume and US stock returns.

	rSP500		rSP500		rSP500		rSP500		rSP500(+1)	rSP500(+2)	rSP500(+3)	rSP500(+4)	
	-1		-2		−3		−4		−5	−6	−7	−8	
Constant	−0.031		−0.036		−0.049		−0.227		−0.524	−0.267	−0.334	−0.534	
	(0.240)		(0.201)		(0.221)		(0.310)		(0.377)	(0.274)	(0.322)	(0.398)	
ΔGSV	−0.251	***	−0.251	***	−0.239	***			0.118	−0.203	−0.071	−0.081	*
	(0.040)		(0.039)		(0.039)				(0.081)	(0.061)	(0.039)	(0.040)	
ΔGSV(−1)					0.088								
					(0.067)								
ΔGSV(−2)					−0.099								
					(0.043)								
log(Cases)	−0.005				−0.001		−0.015		0.082	0.034	0.054	0.065	
	(0.045)				(0.039)		(0.054)		(0.048)	(0.043)	(0.054)	(0.049)	
Δlog(Cases)			−0.420										
			(0.039)										
ΔEPU	−0.002		−0.001		−0.001		−0.001		−0.003	−0.004	−0.014	−0.016	*
	(0.005)		(0.002)		(0.005)		(0.002)		(0.006)	(0.005)	(0.008)	(0.009)	
ΔADS	0.346		0.267		0.163		0.771		−0.191	−0.018	0.263	−0.037	
	(0.420)		(0.353)		(0.370)		(0.572)		(0.448)	(0.361)	(0.494)	(0.376)	
ΔTED	−8.793	*	−8.958	*	−10.045	*	−5.643		−1.437	−4.636	−4.397	−3.704	
	(5.078)		(4.897)		(6.393)		(5.681)		(4.906)	(7.072)	(5.942)	(9.900)	
rSP500(−1)	−0.384	***	−0.379	***	−0.269	***	−0.392	***	0.276	−0.010	−0.184	0.237	*
	(0.122)		(0.126)		(0.069)		(0.137)		(0.089)	(0.113)	(0.121)	(0.125)	
Adj. R^2	0.369		0.377		0.398		0.188		0.100	0.102	0.056	0.078	
F-Statistic	13.04		13.39		11.10		6.70		3.26	3.30	2.18	2.69	
Obs.	124		124		123		124		123	122	121	120	

Table 4
Regression: relationship between "Coronavirus" search volume and G7 stock returns.

Panel A: domestic search						Panel B: global search			
	Constant	ΔGSV		Adj. R^2	*F*-Statistic		ΔGSV		Adj. R^2
Canada (TSX)	−0.404	−0.188	***	0.219	12.47	Canada	−0.312	***	0.309
	(0.334)	(0.066)					(0.094)		
France (CAC)	−0.044	−0.255	***	0.224	12.83	France	−0.279	***	0.223
	(0.157)	(0.067)					(0.076)		
Germany (DAX)	−0.241	−0.164	***	0.109	6.00	Germany	−0.278	***	0.234
	(0.217)	(0.054)					(0.071)		
Italy (MIB)	0.158	−0.086	**	0.037	2.56	Italy	−0.374	***	0.341
	(0.208)	(0.036)					(0.116)		
Japan (TOPIX)	−0.331	−0.016		0.013	1.53	Japan	−0.102	**	0.073
	(0.232)	(0.020)					(0.044)		
UK (FTSE)	−0.298	−0.196	***	0.221	12.61	UK	−0.236	***	0.202
	(0.166)	(0.051)					(0.065)		
US (SP500)	−0.229	−0.255	***	0.339	21.98	US	−0.272	***	0.312
	(0.269)	(0.045)					(0.073)		

return coefficient implies a 1 standard deviation increase in *GSV* is associated with a return of −5%. In contrast, the coefficient in Table 6 suggests a one standard deviation change in GSV is associated with a stock return of −1.05%.

It is also possible that there are non-linear effects on financial markets resulting from information acquisition via Google searches. This may be one reason why the size of the estimated relationship is reduced following the removal of extreme observations (winsorization). We augment eq. (1B) with a squared *ΔGSV*-term and report the results in Panel B of Table 6. We find some evidence of non-linearity since the linear and quadratic terms are negative and statistically significant in all cases. Positive (negative) changes in *GSV* are associated with negative (positive) returns. For larger increases (decreases) in *GSV* the returns decline (increase) more quickly.

Our analysis focuses on the GSV for the "coronavirus" keyword. The principle reason for this owes to the greater prevalence of searches for this word (Fig. 5). However, we recognise that other search terms are also worthy of investigation and repeat our analysis using alternate search terms. In Table 7 we report the estimated coefficients for stock returns using "COVID" and "COVID-19" as alternate keywords. The sign and statistical significance of the estimates are the same as that reported in Table 4 with the exception of Japan, where the significance is reduced. One possible explanation for this may be that the terms "COVID" and "COVID-19" are less prevalent in Japanese media than is "coronavirus".

3.5. GSV and conditional volatility for G7 stock returns

We have already identified a well-defined relationship between investor attention and stock returns during our sample period. We now examine the effect of investor attention on the *volatility* of stock returns in an EGARCH framework. While Da et al. (2015) focus on *realised* volatility, we choose to examine *conditional* volatility since there is evidence that the variance of the returns distribution is not constant over the whole sample period. For instance, we have noted a sharp increase in squared returns that can also be discerned from Figure 4. In employing a similar model to investigate currency volatility, Smith (2012) explains that the ARCH (GARCH) effects of Engle (1982) and Bollerslev (1986) are related to the rate at which information flows in the context of the mixture of distributions hypothesis (MDH). Our measure of investor

Table 5
Regression: relationship between "Coronavirus" search interest and G20 stock returns.

Panel A: domestic search						Panel B: global search			
	Constant	ΔGSV		Adj. R^2	*F*-Statistic		ΔGSV		Adj. R^2
Argentina	−0.118	−0.127	**	0.030	2.29	Argentina	−0.235	***	0.054
	(0.428)	(0.651)					(0.079)		
Australia	−0.069	−0.218	***	0.217	12.36	Australia	−0.236	***	0.258
	(0.287)	(0.067)					(0.059)		
Brazil	−0.847 *	−0.179	*	0.170	9.40	Brazil	−0.370	***	0.270
	(0.459)	(0.109)					(0.120)		
China	−0.014	−0.013		0.004	1.15	China	−0.017		0.004
	(0.171)	(0.031)					(0.035)		
India	−0.559	−0.053		0.038	2.62	India	−0.107		0.032
	(0.324)	(0.050)					(0.080)		
Indonesia	−0.483	−0.015	*	0.036	2.51	Indonesia	−0.044		0.023
	(0.227)	(0.008)					(0.063)		
Mexico	−0.381	−0.014		−0.008	0.68	Mexico	−0.133	***	0.078
	(0.235)	(0.020)					(0.036)		
Russia	−0.316	−0.101	*	0.048	3.07	Russia	−0.171	***	0.123
	(0.362)	(0.052)					(0.058)		
Saudi Arabia	−0.419	−0.026		−0.002	0.92	Saudi Arabia	−0.105	***	0.041
	(0.229)	(0.042)					(0.037)		
South Africa	−0.407	−0.050		0.042	2.80	South Africa	−0.202	***	0.136
	(0.298)	(0.043)					(0.070)		
South Korea	0.211	−0.022		−0.009	0.62	South Korea	−0.107	**	0.032
	(0.266)	(0.019)					(0.043)		
Turkey	−0.558	−0.032		0.044	2.91	Turkey	−0.177	***	0.168
	(0.415)	(0.028)					(0.048)		

Note: This table reports the estimated coefficients for the regression specified in Eq. (1B). The dependent variable is the daily stock index return in excess of the risk free rate. The explanatory variable is the daily change (first difference) in the Google Search Volume (*ΔGSV*) for "Coronovirus" in the specified country (Panel A) or globally (Panel B). Coefficient estimates for the control variables (which include the natural log of the number of new COVID-19 cases) are not reported. Newey-West standard errors are shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level respectively. Sample period: 31-Dec-19 - 30-Jun-20

Table 6
Robustness Tests: "Coronavirus" search volume and G7 stock returns.

Panel A: winsorized, deseasonalized, and standardized GSV				Panel B: non-linear effect of GSV on G7 stock returns				
	ΔGSV		Adj. R^2	ΔGSV		ΔGSV^2		Adj. R^2
Canada	−1.182	***	0.232	−0.233	***	−0.009	***	0.359
	(0.443)			(0.054)		(0.002)		
France	−1.094	***	0.160	−0.210	***	−0.008	***	0.271
	(0.323)			(0.039)		(0.001)		
Germany	−1.104	***	0.161	−0.230	***	−0.007	***	0.265
	(0.312)			(0.043)		(0.001)		
Italy	−1.433	***	0.238	−0.264	***	−0.013	***	0.444
	(0.473)			(0.053)		(0.001)		
Japan	−0.393	**	0.038	−0.095	**	−0.003	**	0.061
	(0.195)			(0.044)		(0.002)		
UK	−0.884	***	0.131	−0.176	***	−0.007	***	0.246
	(0.289)			(0.037)		(0.001)		
US	−1.046	***	0.257	−0.212	***	−0.007	***	0.338
	(0.341)			(0.044)		(0.002)		

attention, *GSV*, may be thought of as an information flow gauge. Table 2 characterizes returns as skewed and with fat-tails (leptokurtic). The EGARCH framework of Nelson (1991) is well suited to modelling such returns and allows for the asymmetric effects of news. To account for the leptokurtic distribution of returns we allow the error term to follow a Generalized Error Distribution (GED). We adopt the following specification to model the conditional volatility of returns:

$$r_t = \delta_0 + \delta_1 \Delta \mathrm{GSV}_t \ (2A)$$

$$log\left(\sigma_t^2\right) = \lambda_0 + \lambda_1 \frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} + \lambda_2 \sigma_{t-1}^2 + \gamma_E \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \lambda_3 \Delta \mathrm{GSV}_t \ (2B)$$

where asymmetry is present if $\gamma_E \neq 0$ and ΔGSV_t is the daily change (first difference) in the domestic Google search volume of the "coronavirus" key word. Table 8 reports the estimated results with Huber-White standard errors.

The mean equation for G7 stock returns confirms the negative relationship between changes in *GSV* and returns. Owing to the positive and significant λ_1 coefficient, the estimated conditional variance equation suggests that there is a positive relationship between past and current levels of variance (ARCH effect). λ_2 close to 1 and highly significant indicates a high degree of volatility persistence across markets. Only the US appears to have a statistically significant asymmetric response, with $\gamma_E < 0$ implying that bad news will increase volatility more than good news of the same magnitude. Most importantly for the purposes of our study, increases in investor attention (*GSV*) are positively related to conditional volatility and statistically significant. That is, higher investor attention is associated with greater volatility for all G7 stock markets. This is consistent with the MDH whereby an increase in the rate of information arrival owing to more Google searches serves to increase volatility (and trading volume).

Estimates for conditional variance of G7 stock returns are illustrated in Fig. 6. In most cases the peak is found in mid-March, coinciding with the enactment of lockdown measures in the US and elsewhere, with

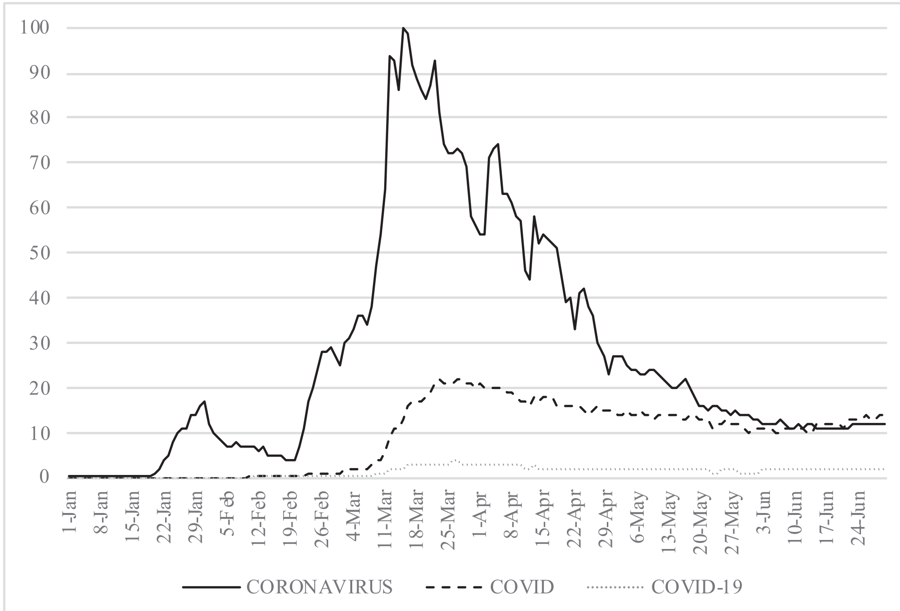

Fig. 5. Google search volume for alternate search terms (Global).

Table 7
Robustness: Importance of alternate search terms on G7 stock returns.

	ΔGSV COVID		Adj. R^2		ΔGSV COVID-19		Adj. R^2
Canada (TSX)	−0.408	***	0.344	Canada	−0.381	***	0.310
	(0.084)				(0.114)		
France (CAC)	−0.312	***	0.181	France	−0.303	***	0.172
	(0.080)				(0.108)		
Germany (DAX)	−0.322	***	0.186	Germany	−0.339	***	0.214
	(0.080)				(0.095)		
Italy (MIB)	−0.349	***	0.194	Italy	−0.353	**	0.201
	(0.123)				(0.146)		
Japan (TOPIX)	−0.088		0.036	Japan	−0.089		0.038
	(0.060)				(0.068)		
UK (FTSE)	−0.291	***	0.202	UK	−0.272	***	0.179
	(0.064)				(0.087)		
US (SP500)	−0.375	***	0.359	US	−0.308	**	0.286
	(0.071)				(0.132)		

direct market intervention by central banks, and with a jump in squared returns. Conditional variance for Japanese stocks has additional sharp peaks in late February, early April, and mid-June, consistent with sharp increases of COVID-19 cases, accompanied by a spike in *GSV*, in Japan at those times.

3.6. Extended sample period

To demonstrate that the identified effect of investor attention is not limited solely to our relatively short sample period, we perform an additional analysis for an extended sample period. Unfortunately, searches for the "coronavirus" keyword seem to be relatively unique to the COVID-19 crisis. To overcome this issue, we identify "pandemic" as a keyword that has the attributes of association with the COVID-19 crisis along with prior health crises, and unlike the term "virus" is not associated with computer code.

We obtain a weekly *GSV* time series for global searches of the "pandemic" keyword from Google Trends for the period running from January 2006 to June 2020. Fig. 7 shows that this peaks in March 2020, but there is also a number of other small increases during the sample period, and a spike in the spring of 2009 (approximately 20% of the March 2020 level). This spike in *GSV* coincides with the emergence of the H1N1 (swine flu) virus.[19]

Adopting the same EGARCH specification, we find results (Table 9) that are broadly consistent with those already reported. Global *GSV* for "pandemic" is negatively associated with stock returns and positively associated with conditional variance for the longer sample period. This provides additional evidence that investor attention plays an important role in stock market dynamics. There is also confirmation of volatility persistence in addition to greater evidence of asymmetry and ARCH effects across all markets.

3.7. Bond markets

Finally, we examine whether government bond yields are also influenced by investor attention during this extraordinary period. Considering the daily change (first-difference) in government bond yields (Table 10) we see that the influence of investor attention is much more limited. *GSV* only has a significant relationship with Italian 10-year yields, and the identified relationship is positive, contrary to the relationship found for stock markets. This suggests that investor attention sparked concern about the likelihood of an Italian default. The lower prevalence of well-defined relationships in government bond markets points to a clientele effect whereby a smaller portion of trading activity is conducted by retail investors. This also lends support to the idea that *GSV* is primarily a measure of investor attention among retail investors. It is also possible that the response identified in the bond market is more directly related to the central bank intervention in money and credit markets during the crisis.

While there appears to be a high degree of persistence in volatility across G7 bond yields, the ARCH effect only seems to be significant for Italian yields. Interestingly, while investor attention does not have a well-defined relationship with yield changes, it does appear to positively influence the conditional variance of bond yields.

[19] https://www.cdc.gov/flu/pandemic-resources/2009-h1n1-pandemic.html

Table 8
EGARCH: effect of "Coronavirus" search interest on conditional volatility in G7 stock returns.

	Canada	France	Germany	Italy	Japan	UK	US
Mean							
Constant	0.271***	0.018	0.004	0.135**	−0.149	0.078	0.048**
	(0.061)	(0.168)	(0.043)	(0.062)	(0.145)	(0.144)	(0.023)
ΔGSV	−0.070**	−0.196***	−0.184***	−0.055**	−0.008	−0.119***	−0.204***
	(0.031)	(0.052)	(0.035)	(0.025)	(0.017)	(0.045)	(0.067)
Conditional variance							
Constant	−1.561***	0.148***	0.080***	0.248	−0.122	0.116	0.342*
	(0.267)	(0.001)	(0.012)	(0.175)	(0.272)	(0.369)	(0.176)
$\lvert\varepsilon_{t-1}\rvert / \sigma_{t-1}$	0.303***	0.518***	0.388***	0.313	0.836***	0.168***	0.405**
	(0.043)	(0.143)	(0.002)	(0.220)	(0.238)	(0.031)	(0.189)
$\log(\sigma^2_{t-1})$	0.949***	0.921***	0.983***	0.978***	0.467*	0.862***	0.972***
	(0.063)	(0.055)	(0.020)	(0.002)	(0.247)	(0.135)	(0.000)
$\varepsilon_{t-1} / \sigma_{t-1}$	−0.374*	−0.236	−0.287*	−0.127	−0.109	−0.290	−0.434***
	(0.201)	(0.145)	(0.154)	(0.088)	(0.195)	(0.256)	(0.080)
ΔGSV	0.058**	0.038***	0.059***	0.048***	0.033*	0.031***	0.040***
	(0.026)	(0.006)	(0.010)	(0.007)	(0.015)	(0.015)	(0.015)
Adj. R^2	0.108	0.219	0.118	0.020	0.010	0.187	0.172
Obs.	124	124	124	124	124	124	124

Note: This table presents estimated coefficients for the effect of "Coronavirus" search interest on the conditional volatility of G7 stock returns, modelled using EGARCH. The dependent variables are daily stock index returns in excess of the risk free rate. In addition to the EGARCH terms, the explanatory variable is the daily change (first difference) in the Google Search Volume (*ΔGSV*) for "Coronovirus" in the specified country. Huber-White standard errors are shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level respectively. Sample period: 31-Dec-19 - 30-Jun-20

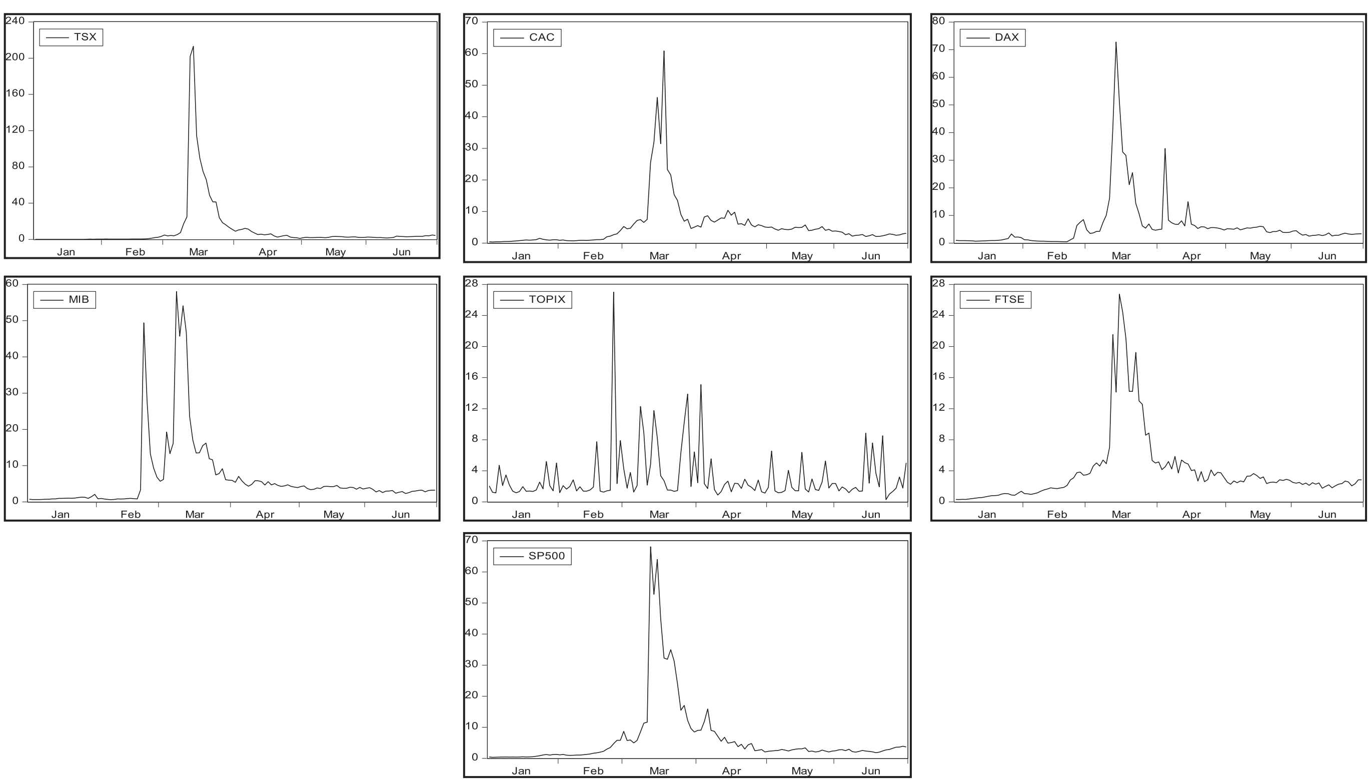

Fig. 6. Conditional Variance G7 stock markets (H1 2020).

4. Conclusion

Information is more quickly reflected in asset prices when investors pay more attention to news events. The literature has shown that this elicits an increase in trading activity along with a price response and greater volatility. The COVID-19 crisis provides a unique setting in which to examine the market response to investor attention during a period of extreme uncertainty.

Our empirical results add to the investor attention literature. Utilising Google search volume (*GSV*) as a proxy for investor attention we show that there is an economically and statistically significant relationship with global stock market returns. A relationship that remains even after controlling for the number of COVID-19 cases and macroeconomic effects. Generally, our results indicate that *GSV* is a proxy for the attention of retail investors and illustrate that stock returns are lower, and price volatility is higher, when investor attention rises. The results suggest that retail investors are most likely conducting online searches for information to resolve household uncertainty during the COVID-19 crisis.

The results have implications for retail investors seeking to trade

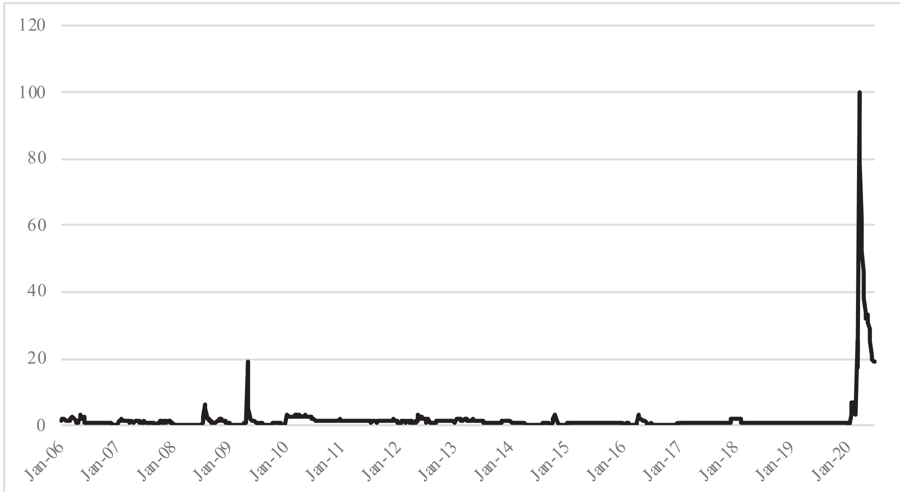

Fig. 7. Google search volume for “Pandemic” (global).

Table 9
EGARCH: Effect of “pandemic” search interest on conditional volatility in G7 stock returns (extended sample period).

	Canada		France		Germany		Italy		Japan		UK		US	
Mean														
Constant	0.031		−0.033		0.077		−0.080		0.054		−0.039		0.151	**
	(0.068)		(0.095)		(0.099)		(0.109)		(0.094)		(0.074)		(0.063)	
ΔGSV	−0.394	***	−0.220	**	−0.145	*	−0.289	***	−0.171	**	−0.181	***	−0.178	***
	(0.071)		(0.086)		(0.077)		(0.107)		(0.075)		(0.055)		(0.054)	
Conditional variance														
Constant	0.107	***	0.072	**	0.109	**	0.041	*	0.123	**	0.015		0.128	***
	(0.031)		(0.033)		(0.054)		(0.022)		(0.059)		(0.029)		(0.044)	
$\lvert\varepsilon_{t-1}\rvert / \sigma_{t-1}$	0.269	***	0.072	**	0.238	***	0.037	*	0.277	***	0.152	***	0.406	***
	(0.047)		(0.034)		(0.056)		(0.023)		(0.044)		(0.030)		(0.046)	
$\log(\sigma^2_{t-1})$	0.917	***	0.934	***	0.851	***	0.968	***	0.822	***	0.914	***	0.861	***
	(0.015)		(0.013)		(0.028)		(0.008)		(0.030)		(0.013)		(0.022)	
$\varepsilon_{t-1} / \sigma_{t-1}$	−0.174	***	−0.237	***	−0.247	***	−0.140	***	−0.199	***	−0.275	***	−0.243	***
	(0.026)		(0.027)		(0.025)		(0.018)		(0.028)		(0.022)		(0.031)	
ΔGSV	0.055	***	0.018		0.028	**	0.019		0.041	***	0.036	***	0.042	**
	(0.017)		(0.017)		(0.012)		(0.018)		(0.015)		(0.012)		(0.017)	
Adj. R^2	0.070		0.065		0.053		0.075		0.042		0.063		0.026	
No. Obs.	750		750		750		750		750		750		750	

Note: This table presents estimated coefficients for the effect of “Pandemic” search interest on the conditional volatility of G7 stock returns, modelled using EGARCH. The dependent variables are weekly stock index returns in excess of the risk free rate. The explanatory variable is the weekly change (first difference) in the global Google Search Volume (*ΔGSV*) for “Pandemic”. Huber-White standard errors are shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level respectively. Sample period: Jan-2006 - Jun-2020

Table 10
EGARCH: Effect of “Coronavirus” search interest on conditional volatility in G7 government bond yields.

	Canada		France		Germany		Italy		Japan		UK		US	
Mean														
Constant	−0.019	**	−0.010	**	−0.011	***	0.005		0.000		−0.013	***	−0.994	**
	(0.007)		(0.004)		(0.003)		(0.017)		(0.002)		(0.004)		(0.477)	
ΔGSV	−0.087	*	0.037		−0.078		0.137	**	−0.004		−0.107		−0.225	
	(0.050)		(0.127)		(0.054)		(0.063)		(0.011)		(0.074)		(0.226)	
Conditional variance														
Constant	−0.518		−0.136		0.280		−0.550		−0.842		−0.514		−0.068	
	(0.508)		(0.089)		(0.254)		(0.884)		(0.512)		(0.695)		(0.220)	
$\lvert\varepsilon_{t-1}\rvert / \sigma_{t-1}$	0.067		0.133		0.094		0.634	***	0.422		0.394		0.034	
	(0.265)		(0.147)		(0.312)		(0.160)		(0.291)		(0.360)		(0.294)	
$\log(\sigma^2_{t-1})$	0.918	***	0.937	***	0.933	***	0.971	***	0.935	***	0.936	***	0.946	***
	(0.063)		(0.000)		(0.000)		(0.000)		(0.049)		(0.026)		(0.000)	
$\varepsilon_{t-1} / \sigma_{t-1}$	−0.258	*	−0.091		−0.194		−0.052		−0.132		−0.114		−0.239	**
	(0.138)		(0.158)		(0.185)		(0.110)		(0.116)		(0.261)		(0.118)	
ΔGSV	0.029	*	0.056	***	0.063	***	0.053	***	0.033	*	0.112	***	0.055	***
	(0.017)		(0.011)		(0.013)		(0.016)		(0.018)		(0.024)		(0.016)	
Adj. R^2	0.000		0.017		0.028		0.020		−0.004		0.012		0.015	

Note: This table presents estimated coefficients for the effect of “Coronavirus” search interest on the conditional volatility of G7 bond yields, modelled using EGARCH. The dependent variables are daily changes in bond yields. In addition to the EGARCH terms, the explanatory variable is the daily change (first difference) in the domestic Google Search Volume (*ΔGSV*) for “Coronovirus”. Huber-White standard errors are shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level respectively. Sample period: 31-Dec-19 - 30-Jun-20 optimally and suggest that caution should be taken in adding to portfolio risk during periods with high levels of investor attention. There may also be implications for policy makers and suggest that monitoring of Google search volume may provide warning of an increased amount of losses among retail investors during periods of high volatility.

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Declaration of Competing Interest

None.

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